Answers and marking guidance

Award one mark per correct, developed point unless stated. Other valid answers should be credited.

2 State what is meant by a member in tension.

- The member is being pulled / stretched by the forces at its ends

3 Quick check

- a) Strut
- b) 1.0 m
- c) CWM = ACWM
- d) 980 N

4 Key terms

- 1 – D
- 2 – F
- 3 – B
- 4 – E
- 5 – C
- 6 – A

5 State the two conditions of equilibrium.

- The sum of the forces = 0 (upward = downward, left = right)
- The sum of the moments = 0 (clockwise = anticlockwise)

6 Calculate each reaction.

- Total load = 2 × 6 = 12 kN acting at the centre
- Each reaction = 6 kN

7 Calculate the magnitude and direction of the resultant.

- $R = \sqrt{(400^2 + 300^2)} = 500$ N
- $\theta = \tan^{-1}(300 \div 400) = 36.9°$
- Direction stated relative to the horizontal